Industrial Internship Report on

” Full-Stack Student Management System”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. This report presents the development and implementation of a Full-Stack Student Management System developed during the Free Summer Internship in Python organized by upskill Campus in collaboration with The IoT Academy and UniConverge Technologies Pvt Ltd (UCT). The objective of this project was to design a system capable of managing student records efficiently using modern software development practices. The system allows users to add, update, delete, search, and manage student information through an interactive web interface and a Python backend. The project was developed using Python, Flask, Object-Oriented Programming principles, JSON-based data storage, and RESTful APIs. Various validation mechanisms were implemented to ensure data integrity and reliability. This internship provided valuable exposure to software development, project planning, testing, documentation, and deployment using GitHub. It enhanced my technical skills and understanding of real-world application development. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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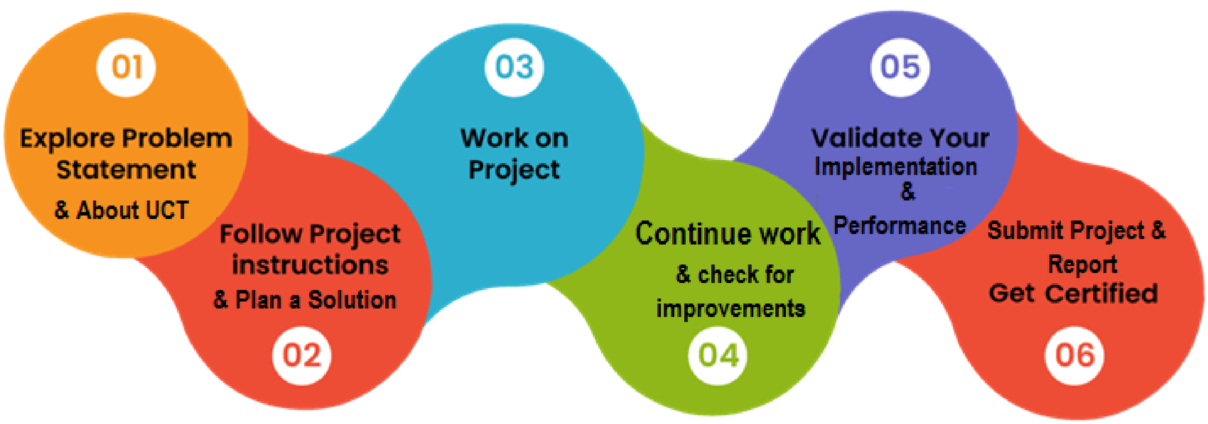
1 Preface

The six-week internship program provided an excellent opportunity to gain practical exposure to Python programming and software development. Throughout the internship, I learned various programming concepts including Object-Oriented Programming, File Handling, Data Validation, Web Development using Flask, and GitHub version control.

As part of the internship project, I developed a Full-Stack Student Management System that helps educational institutions manage student information efficiently. The project involved planning, designing, implementing, testing, and documenting a complete software solution.

This internship improved my technical knowledge, problem-solving ability, project management skills, and understanding of industry practices. I would like to express my sincere gratitude to upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd, and all mentors who supported me during this internship.

I encourage future students to actively participate in internships as they provide valuable practical experience and industry exposure.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

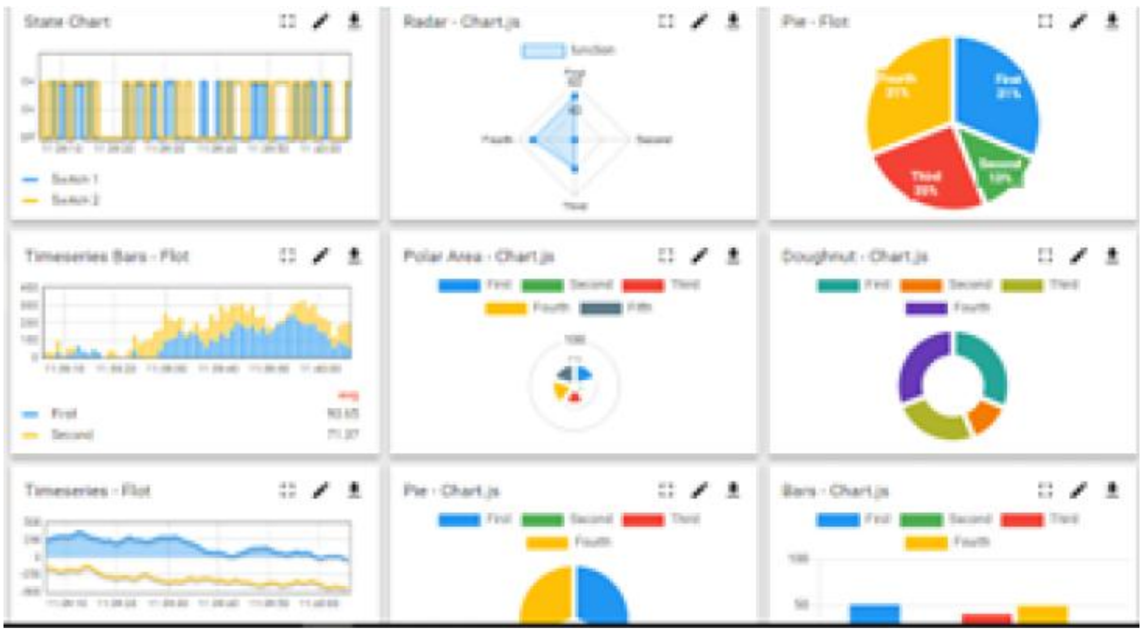
i. UCT IoT Platform (uct Insight)

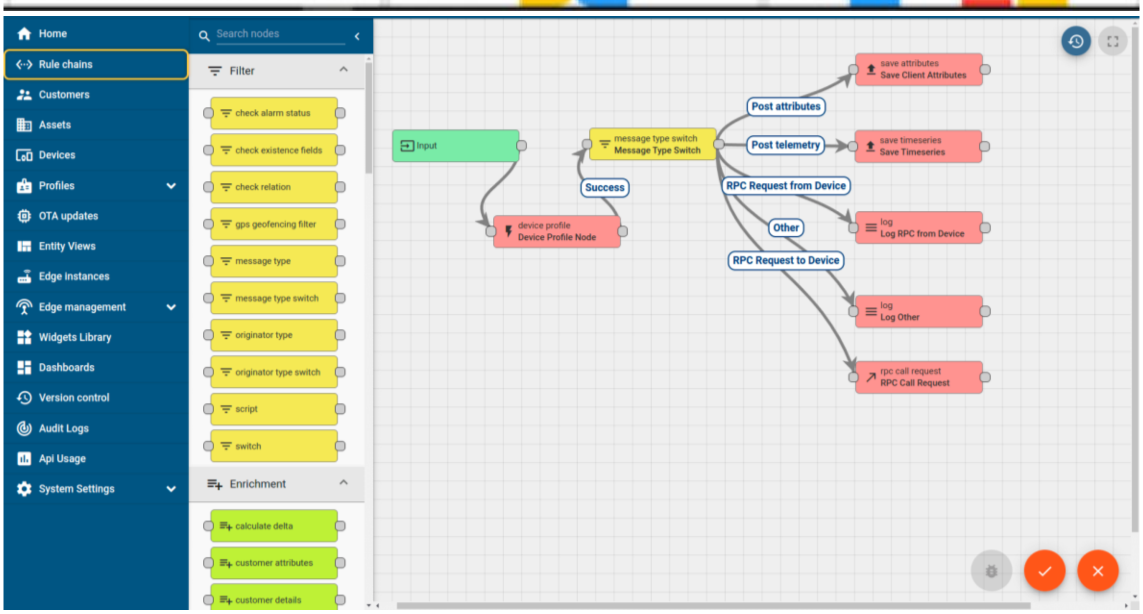
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

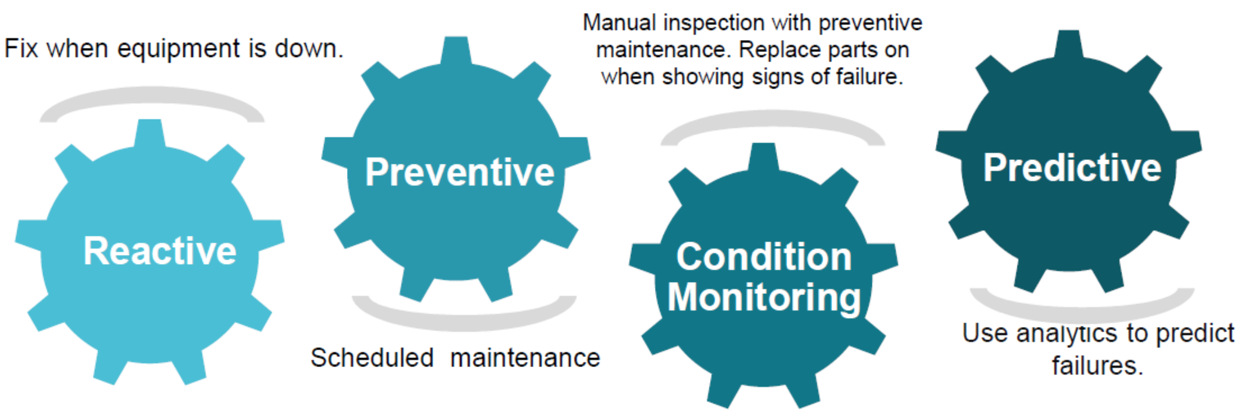
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

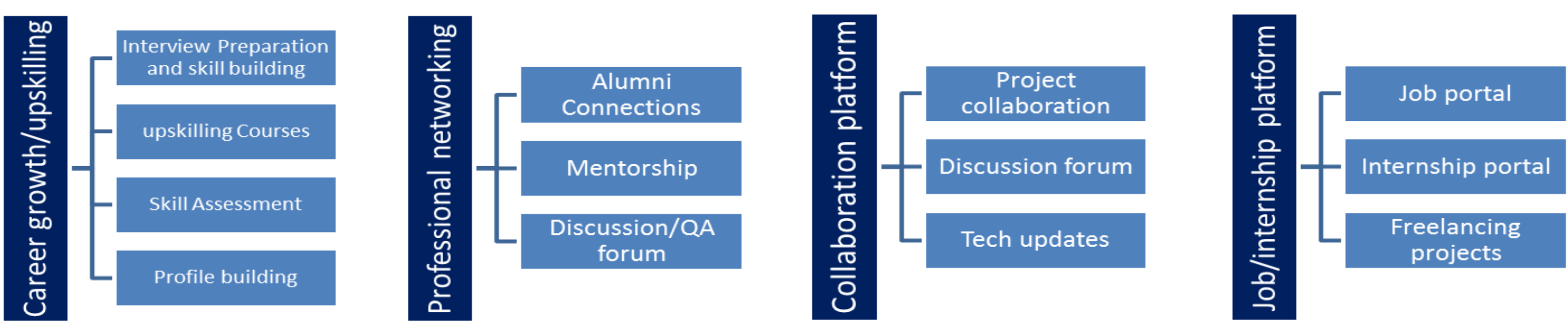
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Python Official Documentation - https://docs.python.org

[2] Flask Documentation - https://flask.palletsprojects.com

[3] GitHub Documentation - https://docs.github.com

2.6 Glossary

Terms	Acronym
OOP	Object Oriented Programming
API	Application Programming Interface
JSON	JavaScript Object Notation
CRUD	Create Read Update Delete
SMS	Student Management System

3 Problem Statement

In the assigned problem statement

Educational institutions often face difficulties in maintaining student records efficiently. Traditional paper-based systems and spreadsheets are prone to data redundancy, inconsistency, and human errors. Searching and updating records can be time-consuming and inefficient.

The objective of this project is to develop a Full-Stack Student Management System that provides an efficient, secure, and user-friendly platform for managing student information. The system should allow administrators to add, update, search, view, and delete student records while maintaining data integrity through proper validation mechanisms.

4 Existing and Proposed solution

Criteria	Existing (Paper/Spreadsheet)	Proposed Full-Stack Web App
Data Integrity	High risk of typing errors, format discrepancies.	Automated two-tier validation (inline browser validation and robust backend constraints).
Search Speed	Manual lookup, scanning entire rows ($O(N)$).	Hash-map based primary-key lookup ($O(1)$) and instant client-side text filtering.
Duplicate Prevention	Relies on visual double-checks.	Strict primary key validation rejecting duplicate IDs.
User Experience	Monotonous grid views, easy to make mistakes.	Beautiful, glassmorphic single-page web dashboard with interactive cards, instant sort toggles, and toast updates.
Portability	Hard to move spreadsheets across platforms.	Cross-platform Python-Flask backend storing records in light JSON files.

4.1 Code submission (Github link) : https://github.com/Kusal76/upskillcampus

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

The updated architecture routes web traffic through a Flask middleware layer, while CLI users interact directly via the main executable loop. Both interfaces utilize the same core manager and persist data in students.json.

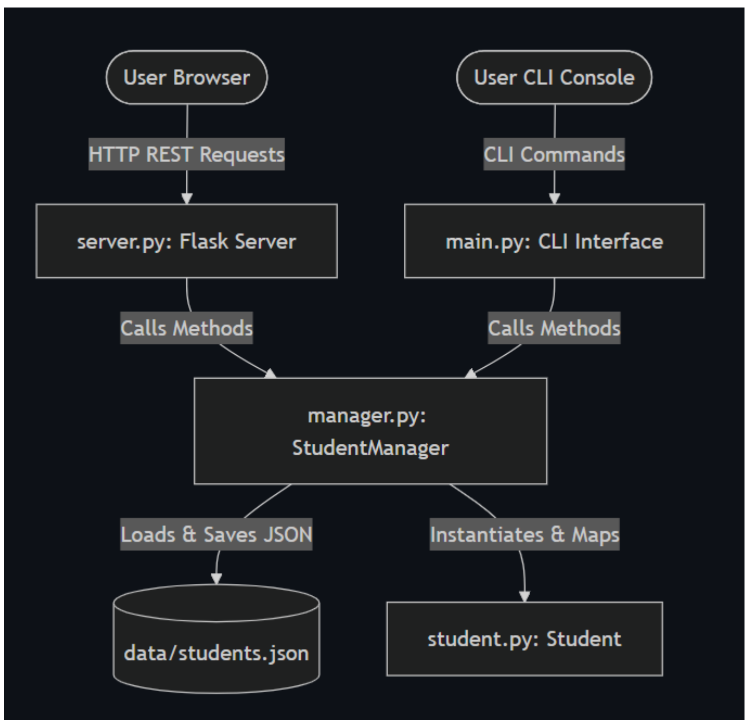
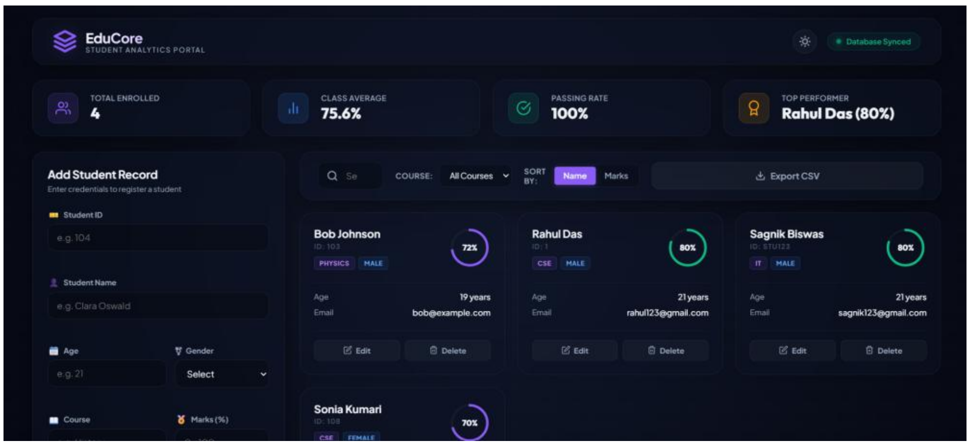


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.



6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

Test ID	Area Tested	Input Provided	Expected Result	Status
TC01	UI Load	Open browser at http://127.0.0.1:5000/	Serves the dark-mode dashboard; loads and displays pre-seeded students in cards.	**PASSED**
TC02	Add Student (Valid)	ID: 1004, Name: David, Age: 21, Gender: Male, Course: Physics, Marks: 89.0, Email: david@edu.com	Renders a new card instantly; displays success toast; writes to students.json.	**PASSED**
TC03	Duplicate ID Alert	Input ID: 1001 (already exists)	Inline client error: This Student ID already exists. Form submit blocked.	**PASSED**

Test ID	Area Tested	Input Provided	Expected Result	Status
TC04	Invalid Email	Email: david_at_gmail (no .com)	Inline error: Invalid email address format. Form submit blocked.	**PASSED**
TC05	Age Out-of-bounds	Age: -2	Inline error: Age must be greater than 0. Form submit blocked.	**PASSED**
TC06	Marks Boundary	Marks: 102	Inline error: Marks must be between 0 and 100. Form submit blocked.	**PASSED**
TC07	Search Filter	Type "Alice" in search box	Instantly filters the visible grid to show only Alice's card.	**PASSED**
TC08	Edit Record	Click Edit on ID 1003 -> Modify Marks to 90 -> Click Update	Grid updates in-place; ID field is locked (readonly); saves to students.json.	**PASSED**
TC09	Sort Grid	Click "Sort by Marks"	Grid cards instantly sort descending with highest marks on top.	**PASSED**
TC10	Delete Record	Click Delete on ID 1001 -> Click Confirm	Card fades out; class average and student counts update; record deleted from file.	**PASSED**

6.2 Test Procedure

- Launch application.
- Add new student records.
- Validate duplicate IDs.
- Validate email format.
- Validate marks range.
- Search existing records.
- Update student information.
- Delete records.
- Sort records.
- Verify data persistence.

6.3 Performance Outcome

The Student Management System performed successfully during testing. All CRUD operations worked correctly, and validation mechanisms prevented invalid data entry. The application demonstrated reliable performance and efficient record management.

The system provides fast search operations, smooth user interaction, and accurate data storage using JSON-based persistence.

7 My learnings

During this internship, I gained valuable knowledge and practical experience in:

- Python Programming
- Object-Oriented Programming
- File Handling
- JSON Data Storage
- Flask Web Development
- REST API Development
- Software Testing
- Input Validation Techniques
- GitHub Version Control
- Project Documentation

The internship enhanced my technical knowledge, problem-solving abilities, and software development skills. It also improved my understanding of industry-level project implementation and deployment practices.

8 Future work scope

The project can be enhanced further through the following improvements:

1. Integration with MySQL or PostgreSQL databases.
2. User Authentication and Authorization.
3. Role-Based Access Control.
4. Student Performance Analytics Dashboard.
5. PDF Report Generation.
6. Cloud Deployment Support.
7. Mobile Application Integration.
8. Email Notification System.
9. Data Backup and Recovery Features.
10. Advanced Search and Reporting Modules.

Conclusion

The Full-Stack Student Management System was successfully designed and implemented during the internship period. The project fulfilled all intended objectives and demonstrated the practical application of Python programming, Flask web development, Object-Oriented Programming, and software engineering principles.

The internship provided valuable real-world experience and strengthened my technical foundation for future academic and professional growth.